MCL 316

→ Fundamentals of Mechanical Vibrations

Assignment 1

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P.1)

⇒ Machine weight, $W = 2000$ lb

Force-deflection relation given,

$$F = 2000x + 200x^3$$

Static equilibrium condition,

at equilibrium, $F = W$

$$\therefore 2000 = 2000x + 200x^3$$

$$10 = 10x + x^3$$

$$x^3 + 10x - 10 = 0$$

on solving,

$$x_1 = 0.9217$$

$$x_2, x_3 = -0.46 \pm 3.26i$$

so, only x_1 is real root

$\therefore \boxed{x_1 = 0.9217 \text{ in}}$ (Static equilibrium posn)

Now, equivalent linearized spring constant

$$K_{eq} = \left.\frac{dF}{dx}\right|_{x_1}$$

$$F = 2000x + 200x^3$$

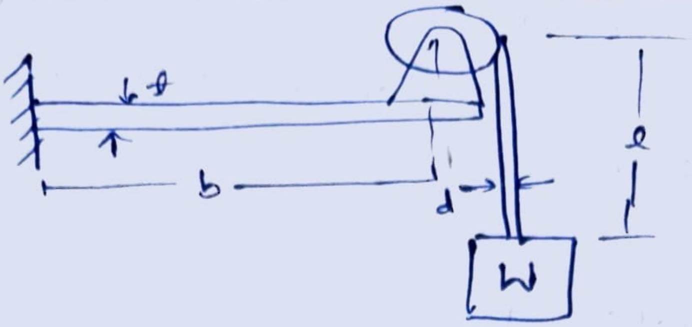
$$\frac{dF}{dx} = 2000 + 600x^2$$

as $x_1 = 0.9217$

$\therefore \boxed{K_{eq} = 2509.71 \text{ lb/in}}$

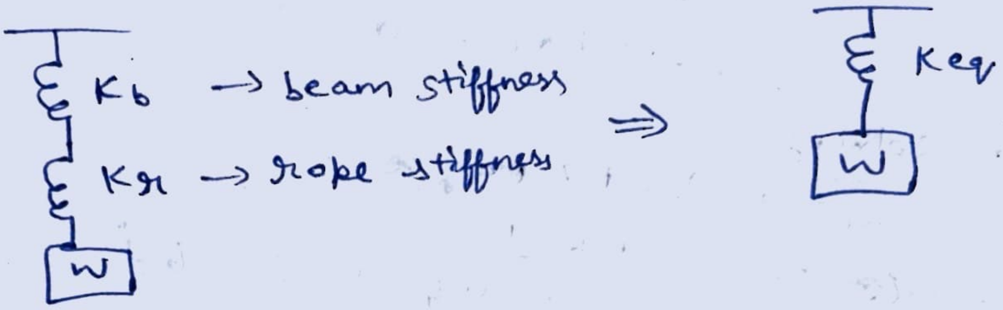
P.2)

⇒



→ The beam and rope deform in series, as the hoisting drum is mounted on a a cantilever beam and the load is suspended by a steel wire rope, thus the total deflection is sum of beam deflection and rope elongation.

→ Youngs modulus of wire and rope = E



① Beam stiffness

→ for cantilever beam carrying load W at free end,

$$\delta_b = \frac{Wb^3}{3EI}$$

$$K_b = \frac{W}{\delta_b}$$

$$\boxed{K_b = \frac{3EI}{b^3}}$$, ~~where,~~ $\cancel{I = \frac{\pi a^4}{64}}$

② Rope stiffness

$$A = \frac{\pi d^3}{4}$$ (Area of rope) (AOC)

extension, $\delta_r = \dfrac{Wl}{AE}$

and ~~for~~ $K_r = \frac{AE}{l}$

$$\boxed{K_r = \frac{\pi d^2 E}{4l}}$$

Now, ~~Keq = Kb~~

Keq (equivalent stiffness)

~~series~~ They are connected in series,

$$\therefore \frac{1}{K_{eq}} = \frac{1}{K_b} + \frac{1}{K_r}$$

$$\frac{1}{K_{eq}} = \frac{b^3}{3EI} + \frac{4l}{\pi d^2 E}$$

$$K_{eq} = \left(\frac{b^3}{3EI} + \frac{4l}{\pi d^2 E}\right)^{-1}$$

~~As beam has~~

Beam has rectangular crossection of width a and thickness t.

$\therefore$ ~~$I = at^3$~~ $\boxed{I = \frac{at^3}{12}}$

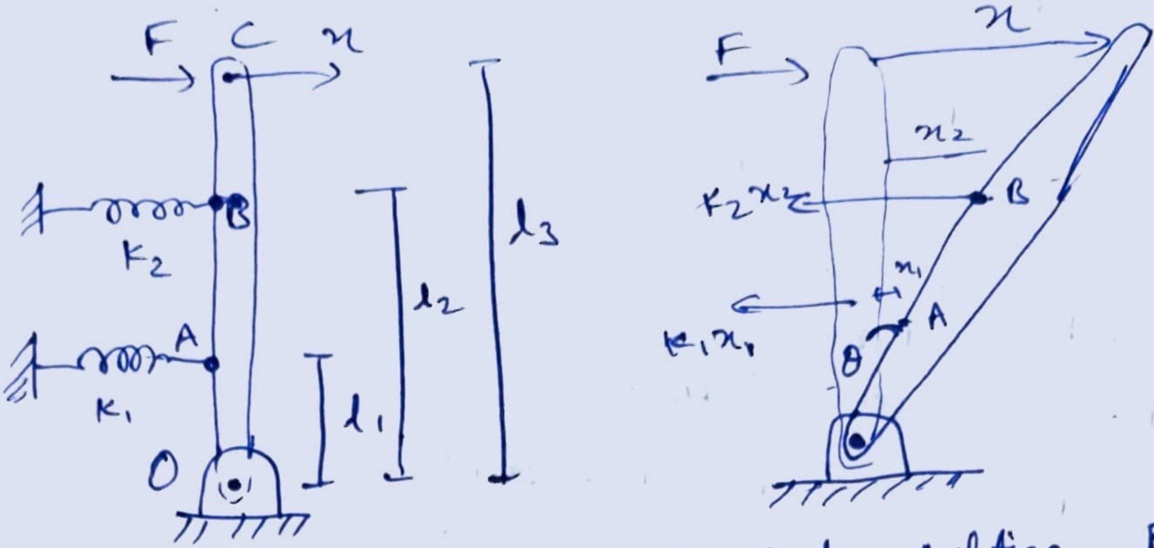
and ~~$K_{eq} = \frac{4b^3}{}$~~

$$\boxed{K_{eq} = \left(\frac{4b^3}{Eat^3} + \frac{4l}{\pi d^2 E}\right)^{-1}}$$

$\therefore$ $\boxed{K_b = \frac{3EI}{b^3}}$ $\boxed{K_r = \frac{\pi d^2 E}{4l}}$

~~$K_{eq} = \frac{4b^3}{Eat^3}$~~ $\boxed{K_{eq} = \left(\frac{b^3}{3EI} + \frac{4l}{\pi d^2 E}\right)^{-1}}$

P.3)



→ To find equivalent spring constant relating F and the horizontal displacement x

for small rotation θ,

Horizontal displacement of spring

→ K_1 : $x_1 = l_1 \theta$

→ K_2 : $x_2 = l_2 \theta$

and Force in spring 1 :

$$F_1 = K_1 x_1 = K_1 l_1 \theta$$

in spring 2 :

$$F_2 = K_2 x_2 = K_2 l_2 \theta$$

Now, moment equilibrium about hinge

→ moment due to applied force $= Fl$

→ moments due to spring forces $= F_1 l_1 + F_2 l_2$

$$\therefore \quad Fl = K_1 l_1^2 \theta + K_2 l_2^2 \theta$$

displacement of top point ⇒ $x = l\theta$

$$\boxed{\theta = \frac{x}{l}}$$

$\therefore$ substituting θ we get

$$F\,l = \left(K_1 l_1^2 + K_2 l_2^2\right)\frac{x}{l}$$

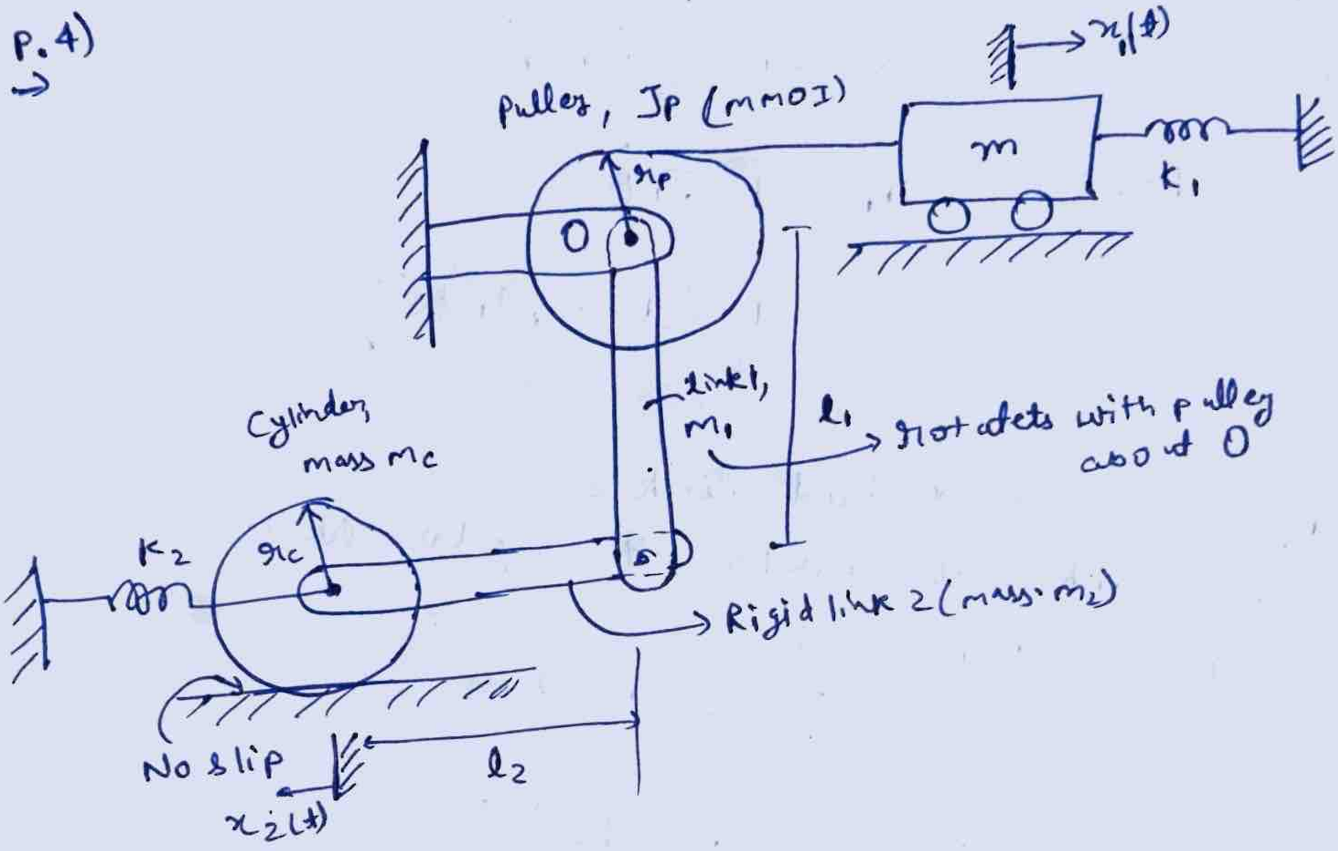
$$F = \frac{K_1 l_1^2 + K_2 l_2^2}{l^2}\,x$$

Comparing with

$$F = K_{eq}\,x$$

$$\therefore \boxed{K_{eq} = \frac{K_1 l_1^2 + K_2 l_2^2}{l^2}}$$

P.4) $\rightarrow$



Taking x_1 as the generalized coordinate.

$\therefore \quad x_1 = r_p\theta$

and $\quad \dot{\theta} = \dfrac{\dot{x}_1}{r_p}$

Kinetic energy of block, $\boxed{T_m = \frac{1}{2} m \dot{x}_1^2}$

Kinetic energy of pulley $T_P = \frac{1}{2} J_P \dot{\theta}^2$

and $T_P = \frac{1}{2} \frac{J_P}{r_P^2} \dot{x}_1^2$ $\left(\text{as } \dot{\theta} = \frac{\dot{x}_1}{r_P}\right)$

→ Kinetic energy of rigid link 1

link 1 rotates with pulley

For a slender rod rotating about one end,

$$J_1 = \frac{1}{3} m_1 l_1^2$$

Hence, $T_1 = \frac{1}{2} J_1 \dot{\theta}^2$

$$= \frac{1}{2}\left(\frac{1}{3} m_1 l_1^2\right) \frac{\dot{x}_1^2}{r_P^2}$$

→ Kinetic energy of rigid link 2

Rigid link 2 also rotates with angular velocity $\dot{\theta}$

for a slendor rod,

$$J_2 = \frac{1}{3} m_2 l_2^2$$

Hence, $T_2 = \frac{1}{2}\left(\frac{1}{3} m_2 l_2^2\right) \frac{\dot{x}_1^2}{r_P^2}$

Now, Kinetic energy of cylinder,

since NO slipping, $v = \dot{x}_2$

also, $\dot{x}_2 = l_2 \dot{\theta} = \frac{l_2 \dot{x}_1}{r_P}$

The cylinder has both translational and rotational kinetic energy.

Translation KE :

$$T_t = \frac{1}{2} m_c v^2$$

Rotation KE : $T_r = \frac{1}{2} J_c \omega^2$

for solid cylinder, $J_c = \frac{1}{2} m_c r_c^2$

and $\omega = \frac{v}{r_c}$

Hence, $T_r = \frac{1}{4} m_c v^2$

Total cylinder energy :

$$T_c = \frac{3}{4} m_c v^2$$

and. $v = \frac{l_2}{r_p} \dot{x}_1$

$$\therefore \; T_c = \frac{3}{4} m_c \left(\frac{l_2}{r_p}\right)^2 \dot{x}_1^2$$

Total kinetic energy

$$T = \frac{1}{2} m_{eq} \dot{x}_1^2$$

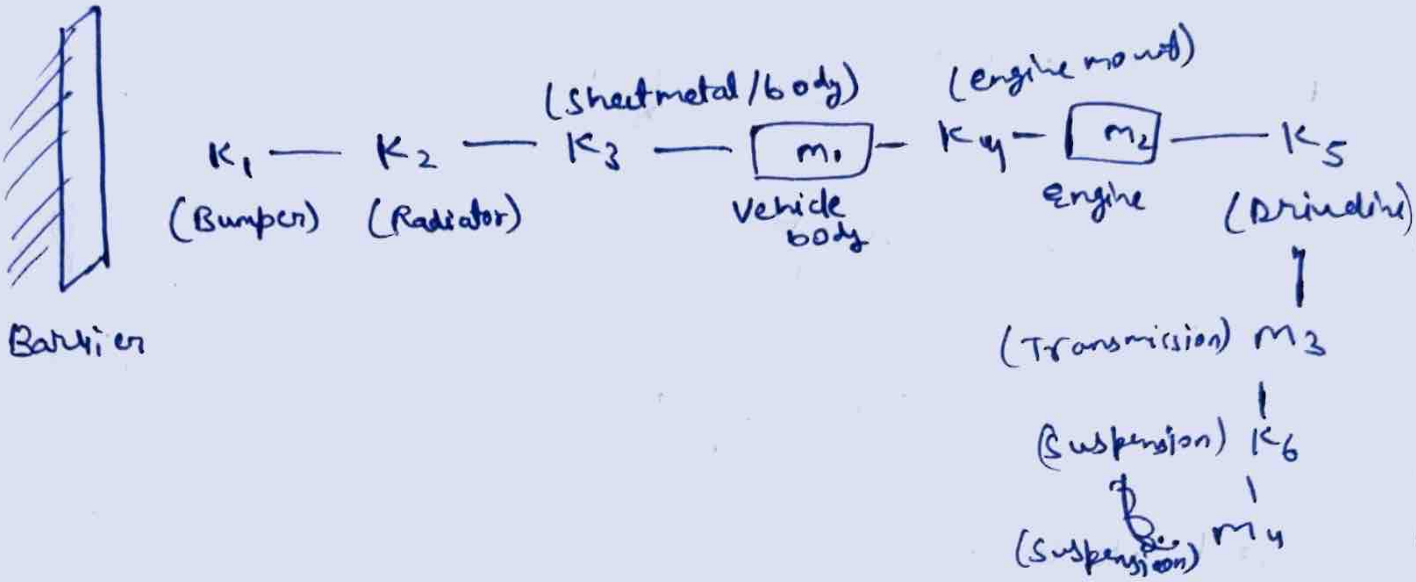
Comparing,

$$\boxed{m_{eq} = m + \frac{J_p}{r_p^2} + \frac{m_1 l_1^2}{3 r_p^2} + \frac{m_2 l_2^2}{3 r_p^2} + \frac{3 m_c l_2^2}{2 r_p^2}}$$

P. 5)

⇒ Assumptions

1. Motion is only in the longitudinal dirⁿ
2. Barrier is perfectly rigid
3. All components behave as lumped masses.
4. Damping is neglected.



Now eqⁿ of motion

Let x_1, x_2, x_3, x_4 be the displacements of the body engine, transmission & suspension, respectively

Now, mathematical model

Body
$$m_1\ddot{x}_1 + (K_1+K_2+K_3+K_4)x_1 - K_4x_2 = 0$$

Engine
$$m_2\ddot{x}_2 + K_4(x_2-x_1) + K_5(x_2-x_3) = 0$$

Transmission
$$m_3\ddot{x}_3 + K_5(x_3-x_2) + K_6(x_3-x_4) = 0$$

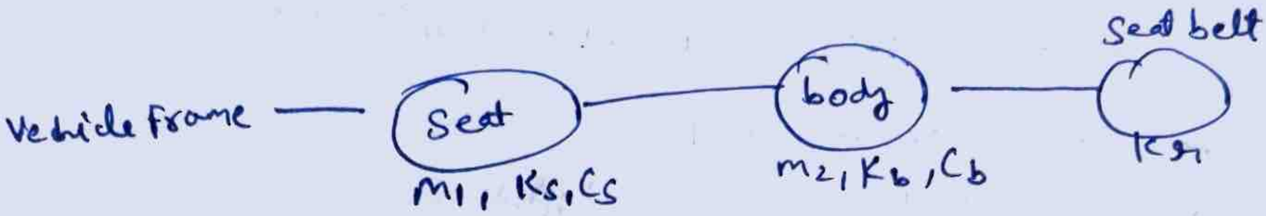
Suspension
$$m_4\ddot{x}_4 + K_6(x_4-x_3) = 0$$

∴ Degree of freedom = 4

P.6)

→ **Assumptions**

1. Motion is only along the direction of impact
2. Human body is represented by lumped masses
3. Seat belt behaves as linear spring
4. Seat cushion & human body behaves as a spring damper.
5. Vehicle structure is rigid.



m — mass
K — stiffness
C — damping

Now,
Let x_1 = seat displacement
x_2 = body displacement

Mathematical model,

Seat,
$$m_1\ddot{x}_1 + C_s\dot{x}_1 + K_s x_1 + C_b(\dot{x}_1 - \dot{x}_2) + K_b(x_1 - x_2) = 0$$

Body,
$$m_2\ddot{x}_2 + C_b(\dot{x}_2 - \dot{x}_1) + K_b(x_2 - x_1) + K_r x_2 = 0$$

$\boxed{DOF = 2}$

P.7)

→ Model 1: Single DOF (SDOF) model

Assumptions
→ Entire vehicle (including passengers) considered as one lumped mass.
→ Suspension is represented by one spring & one damper

m — (K, C) — Road profile $y(t)$

m = Total mass of vehicle
K = Equivalent suspension stiffness
C = equivalent damping coeff

Eqn of motion

$$m\ddot{x} + C(\dot{x} - \dot{y}) + K(x - y) = 0$$

x = vehicle body disp.
$y(t)$ = road disp.

Model 2: Two DOF model

Assumptions
→ Vehicle body and wheel are considered seperately
→ Tyre & suspension have ~~diff stif~~ different stiffness
→ Shock absorber is included

m_s — K_s, C_s — m_u — K_t — Road $y(t)$

m_s = Sprung mass (vehicle body + passengers)
m_u = Unsprung mass (wheel & axle)
K_s = Suspension stiffness
C_s = Shock absorber
K_t = Tyre stiffness

Eq^n of motion

For sprung mass:

$$m_s \ddot{x}_s + C_s(\dot{x}_s - \dot{x}_u) + K_s(x_s - x_u) = 0$$

for unsprung mass:

$$m_u \ddot{x}_u + C_s(\dot{x}_u - \dot{x}_s) + K_s(x_u - x_s) + K_t(x_u - y) = 0$$

Model 3 - Multi DOF (full / refined cars) (4 DOF)

Assumptions

-> The vehicle is represented by four masses

m_1 = Car body

m_2 = Front wheel

m_3 = Rear wheel

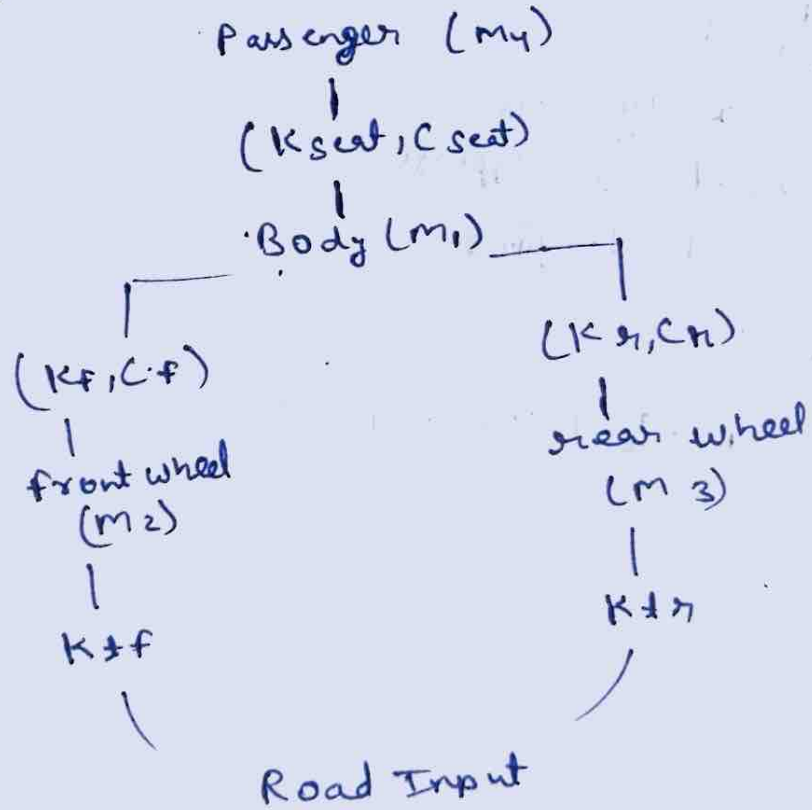
m_4 = Passenger & seat

-> Elastic elements

- front suspension
- Rear suspension
- Seat springs
- Tyres

-> Damping

- Front and rear shock absorbers
- seat damper



Features :

- Includes front & rear suspension
- Consider tyre flexibility
- Includes passenger comfort.

∴

Three mathematicals models are :

1. SDOF - vehicle treated as one mass
2. 2 DOF - body and wheel modelled seperately
3. Multi DOF (4) - body, front wheel, rear wheel & passenger & with suspension, tyres, seats & dampers

P.8)

Model 1: Single DOF (SDOF)

Assumptions :

- Entire human body is considered as one lumped mass
- legs are represented by a spring and damper
- Ground provides the excitation.

Model :

M
|
K, C
|
Ground

m = Total body mass

K = Equivalent leg stiffness

c = Equivalent damping

Eqn of motion

$$m\ddot{x} + c\dot{x} + Kx = F(t)$$

Model 2: Two DOF (2 DOF)

Assumptions

- The body is divided into:
  - m_1 = Upper body (head + torso)
  - m_2 = lower body (hips + legs)

Connected by the spinal column, represented as a spring & damper

Model:

Upper Body (m_1)
|
K_1, C_1
|
Lower Body (m_2)
|
K_2, C_2
|
Ground

features

- Models relative motion b/w upper & lower body
- Suitable for moderate vibration analysis.

Model 3: Four DOF (4DOF)

Masses, upper torso, hips, legs stacked from top to ground, joined by:

- neck
- spinal column
- Abdomen
- legs

Masses

- m_1 = Head
- m_2 = Upper torso
- m_3 = Hips / abdomen
- m_4 = legs

Springs and Dampers

- Neck → K_1, C_1
- Spine → K_2, C_2
- Abdomen → K_3, C_3
- Legs → K_4, C_4

Head (m_1)
|
K_1, C_1
|
Upper Torso (m_2)
|
K_2, C_2
|
Hips (m_3)
|
K_3, C_3
|
Legs (m_4)
|
K_4, C_4
|
Ground

Features

- Captures vibration of individual body segments.
- suitable for studying shock, ride comfort, and biomechanical response.
- Most realistic among the 3 models